

Policy Iteration / Value Iter.

$$s, a \sim \pi(a|s), p(s'|s, a), r(s, a)$$

$$\tau = \{s_0, a_0, r_0, s_1, a_1, r_1, \dots\}$$

$$p(\tau) = p(s_0) \prod_{t=0}^{\infty} p(s_{t+1}|s_t, a_t) \pi(a_t|s_t)$$

$$R_t = r_t + \gamma r_{t+1} + \gamma^2 r_{t+2} + \dots, \gamma \in (0, 1)$$

$$\mathbb{E}_{p(\tau|\pi)} R_0 \rightarrow \max_{\pi}$$

$$V^{\pi}(s) := \mathbb{E}_{p(r|\pi)} [R_t | s_t = s]$$

$$Q^{\pi}(s,a) := \mathbb{E}_{p(r|\pi)} [R_t | s_t = s, a_t = a]$$

$$V^*(s) := \max_{\pi} V^{\pi}(s)$$

$$Q^*(s,a) := \max_{\pi} Q^{\pi}(s,a)$$

$$R_t = r_t + \gamma r_{t+1} + \gamma^2 r_{t+2} + \dots =$$

$$= r_t + \gamma (r_{t+1} + \gamma r_{t+2} + \dots) = r_t + \gamma R_{t+1}$$

$$V^\pi(s_t) := \mathbb{E}[R_t | s_t = s_t] = V^\pi(s_{t+1})$$

$$= \mathbb{E}[r(s_t, a_t) + \gamma R_{t+1} | s_t = s_t] =$$

$$= \mathbb{E}_{\pi(a_t | s_t)} (r(s_t, a_t) + \gamma \mathbb{E}_{p(s_{t+1} | s_t, a_t)} [\mathbb{E}_{a_{t+2}, s_{t+2}, \dots} [R_{t+1} | s_{t+1} = s_{t+1}]])$$

$$V^{\pi}(s) = \mathbb{E}_{\pi(a|s)} \left(r(s,a) + \gamma \mathbb{E}_{p(s'|s,a)} V^{\pi}(s') \right)$$

- Bellman equation for V^{π}

$$Q^{\pi}(s,a) = r(s,a) + \gamma \mathbb{E}_{p(s'|s,a)} \mathbb{E}_{\pi(a'|s')} Q^{\pi}(s',a')$$

$$V^*(s) = \max_{\pi} V^{\pi}(s) \stackrel{\text{def}}{=} \hat{V}^*(s) = \max_{\pi_0(a_0/s_0), \pi_1(a_1/s_1), \dots} V^{\pi_0, \pi_1, \dots}(s)$$

$$= \max_{\pi_0} \left(\mathbb{E}_{\pi_0(a_0/s_0)} \left(r(s_0, a_0) + \gamma \mathbb{E}_{p(s_1/s_0, a_0)} V^{\pi}(s_1) \right) \right)$$

$$= \left\{ \max(a+b, a+c) = a + \max(b, c) \right\} =$$

$$= \max_{\pi_0} \mathbb{E}_{\pi_0(a_0/s_0)} \left(r(s_0, a_0) + \gamma \mathbb{E}_{p(s_1/s_0, a_0)} \left(\max_{\pi_1, \pi_2, \dots} V^{\pi}(s_1) \right) \right)$$

$\hat{V}^*(s_1)$

$$\pi_0^*(a_0 | s_0) = \arg \max_{\pi_0} E_{\pi_0(a_0 | s_0)} y(s_0, a_0) =$$

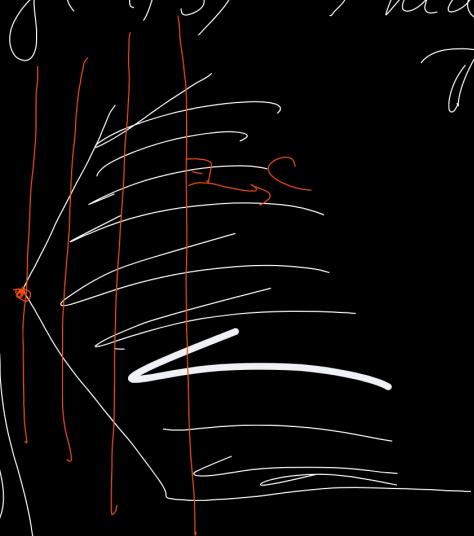
$$= \pi^*(a_0 | s_0)$$

$$\left\{ \begin{array}{l} E_{\pi(a|s)} y(s, a) = \sum_a \pi(a|s) y(a, s) \rightarrow \max_{\pi} \end{array} \right.$$

$$\pi(a|s) \geq 0 \quad \forall a$$

$$\sum_a \pi(a|s) = 1 \quad - \text{LP}$$

$$\pi^*(a|s) = I[a = \arg \max_a y(s, a)]$$



$$V^*(s) = \max_a (r(s,a) + \gamma \mathbb{E}_{p(s'|s,a)} V^*(s'))$$

$$\pi^*(a|s) = \operatorname{argmax}_a (\dots)$$

$$Q^*(s,a) = r(s,a) + \gamma \mathbb{E}_{p(s'|s,a)} \max_{a'} Q^*(s',a')$$

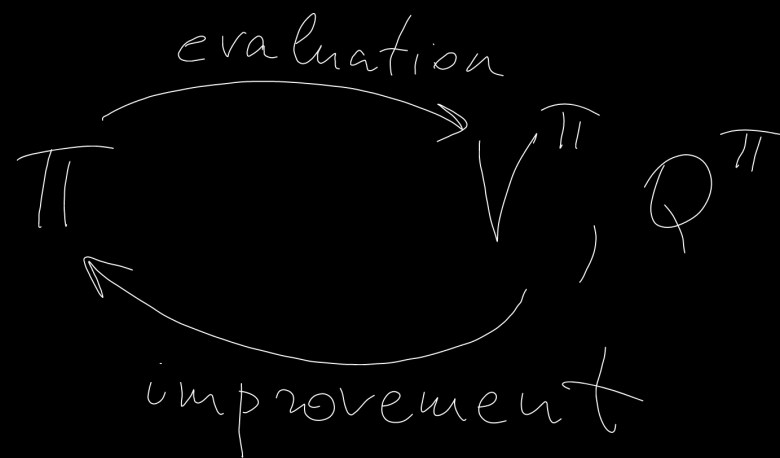
$$\pi^*(a|s) = \operatorname{argmax}_a Q^*(s,a)$$

$$V^{\pi}(s) = \mathbb{E}_{\pi(a/s)} Q^{\pi}(s, a)$$

$$Q^{\pi}(s, a) = r(s, a) + \gamma \mathbb{E}_{p(s'/s, a)} V^{\pi}(s')$$

$$V^{\star}(s) = \max_a Q^{\star}(s, a)$$

$$Q^{\star}(s, a) = r(s, a) + \gamma \mathbb{E}_{p(s'/s, a)} V^{\star}(s')$$



$$V^\pi = F(V^\pi)$$

$$V_{\text{new}}^\pi = F(V_{\text{old}}^\pi)$$

fixed-point iteration

$$\sum_{s'} p(s'/s, a) V^\pi(s')$$

$$V^\pi(s) = \mathbb{E}_{\pi(a|s)} (r(s, a) + \gamma \mathbb{E}_{p(s'|s, a)} V^\pi(s')) \quad \forall s$$

↑
Linear System

Requirement for F :

$$\|F(v) - F(w)\| \leq C \|v - w\| \quad \forall v, w$$

$$\cdot \quad 0 \leq C < 1$$

contraction

$$\|F(v) - F(w)\|_{\infty} \leq C \|v - w\|_{\infty}$$

$$\|x\|_{\infty} := \max(|x_1|, \dots, |x_n|)$$

$$\begin{aligned}
& \text{Fix } s : \left| E_{\pi(a|s)}(\cancel{r(s,a)} + \gamma E_{p(s'|s,a)} V(s')) - \right. \\
& \left. - E_{\pi(a|s)}(\cancel{r(s,a)} + \gamma E_{p(s'|s,a)} W(s')) \right| = \\
& = \gamma \left| E_{\pi(a|s)} E_{p(s'|s,a)} (V(s') - W(s')) \right| \leq \\
& \leq \gamma \sum_{s'} \sum_a p(s'|s,a) \pi(a|s) |V(s') - W(s')| \leq \\
& \leq \gamma \sum_{s'} \sum_a \underbrace{p(s'|s,a) \pi(a|s)}_{p(s'|s)} \|V - W\|_{\infty} = \gamma \|V - W\|_{\infty}
\end{aligned}$$

Iterative Policy Evaluation

Init. $V(s) \forall s$

While True:

$\Delta := 0$

.. for all s :

$\delta = V(s)$

$V(s) = E_{\pi(a|s)}(r(s,a) + \gamma E_{p(s'|s,a)} V(s'))$

$\Delta := \max(\Delta, |\delta - V(s)|)$

if $\Delta \leq \epsilon$, then exit

Policy Improvement

Def. $\hat{\pi} \geq \pi$, iff $V^{\hat{\pi}}(s) \geq V^{\pi}(s) \forall s$

$$\pi, \hat{s}, \hat{\pi}(\hat{s}) = \hat{a}, Q^{\pi}(\hat{s}, \hat{a}) \geq V^{\pi}(\hat{s})$$

$$\hat{\pi}(s) = \pi(s) \quad \forall s \neq \hat{s}$$

$$V^{\pi}(s) = \mathbb{E}_{\hat{\pi}(a|s)} Q^{\pi}(s, a) = \mathbb{E}_{\hat{\pi}(a|s)} \left(r(s, a) + \gamma \mathbb{E}_{p(s'|s, a)} V^{\pi}(s') \right)$$

$$\textcircled{\leq} \dots \leq V^{\hat{\pi}}(s) \Leftrightarrow \hat{\pi} \geq \pi$$

$$\hat{\pi}(a/s) = \arg \max_a Q^{\pi}(s, a) \quad \forall s$$

$$\Rightarrow \hat{\pi} \geq \pi$$

If $\hat{\pi} = \pi$, then $V^{\hat{\pi}}(s) = \max_a \left(r(s, a) + \gamma \mathbb{E}_{p(s'/s, a)} V^{\pi}(s') \right) = V^{\pi}(s) \Rightarrow V^{\hat{\pi}} = V^{\pi}$

Policy Iteration

Init. $\pi(a/s) \forall s$

Step 1. Est. V^π using Iter. Policy Eval.

Step 2. Flag = True

for all s :

$$\delta = \pi(a/s)$$

$$\pi(a/s) = \arg \max_a \left(r(s, a) + \gamma \sum_{s'} P(s'/s, a) V^\pi(s') \right)$$

if $\pi(a/s) \neq \delta$, then Flag = False

If Flag = True, then exit

Value Iteration

Init. $V(s) \forall s$

While True:

$$\Delta = 0$$

for all states s :

$$\bar{v} = V(s)$$

$$V(s) = \max_a \left(r(s, a) + \gamma \mathbb{E}_{p(s'|s, a)} V(s') \right)$$

$$\Delta = \max(\Delta, |\bar{v} - V(s)|)$$

if $\Delta \leq \epsilon$, then exit

Example

reward:

0	0	0	1
0	2	0	-1
0	0	0	0

terminal states

State = position

action $\in \{\leftarrow, \rightarrow, \uparrow, \downarrow\}$

V^0 :

0	0	0	1
0	2	0	-1
0	0	0	0

$\rightarrow$

V^1 :

0	0	0	1
0	2	0	-1
0	0	0	0

$\rightarrow$

V^2 :

0	0	0	1
0	2	0	-1
0	0	0	0

$V^3 \rightarrow V^4 = V^*$

0	0	0	1
0	2	0	-1
0	0	0	0